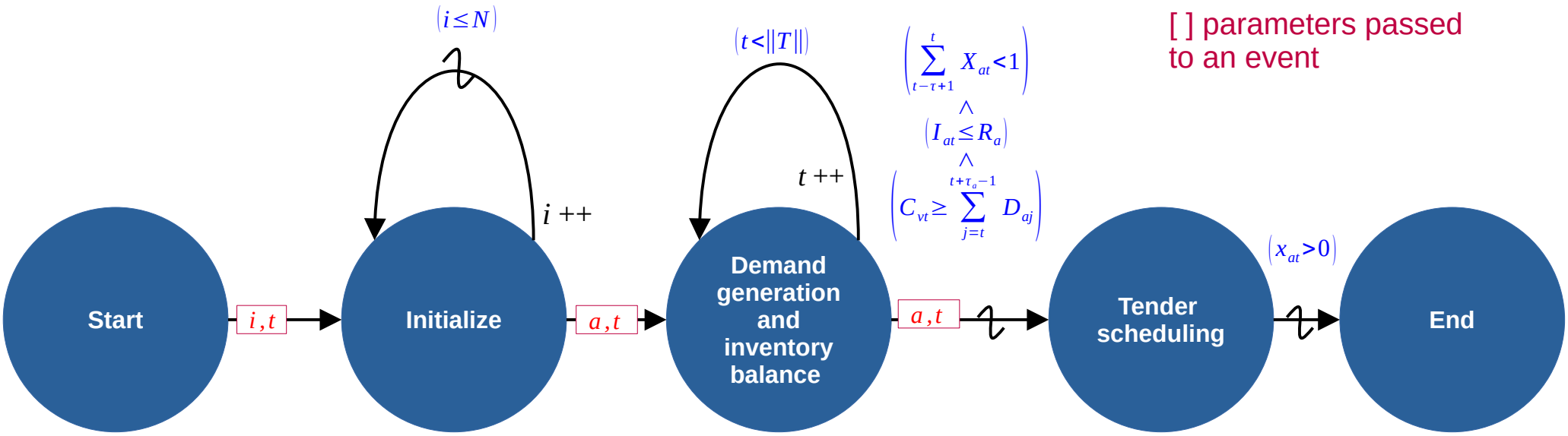
( ) conditions for scheduling events
[ ] parameters passed to an event
$(i \le N)$
$i\ ++$
$(t < \|T\|)$
$t\ ++$
$\left(\sum_{t-\tau+1}^{t} X_{at} < 1\right) \wedge (I_{at} \le R_a) \wedge \left(C_{vt} \ge \sum_{j=t}^{t+\tau_a-1} D_{aj}\right)$
$(x_{at} > 0)$
Start
$i, t$
Initialize
$a, t$
Demand generation and inventory balance
$a, t$
Tender scheduling
End
$i=1,\ t=1$
$a=i$
$u_{a,t-1}=u_{a,o}$
$\delta^q_{at}=k,\ \Delta^q_{vt}=\beta$
$\tau_a=n, R_a=d_a b_t$
$I_{a,t-1}=I_{ao}$
$r_v=m,\ d_a=\delta$
$c_u=c_{uo}, h_{at}=h_{ao}, f_v=f_{vo}$
$C_{vt} \leftarrow rnd\, N(\mu,\sigma)$
$I_{a,t}=I_{a,t-1}+\sum_{v\in V_a}\sum_{q=t-\tau_a+1}^{t}\Delta^q_{vt}-\delta^q_{at}$
$i^q_{a,t}=\delta^q_{a,t}$
$b_t \leftarrow r.v.$
$u_{at}=u_{a,t-1}-i^q_{a,t}+b_t$
$D_{at}=d_a u_{a,t}$
$\omega_{a,t}=u_{at} c_\mu$
$k_{at}=I_{at} h_{at}$
$X_{a,t+\tau_a} \leftarrow 1$
$\sum_{v\in V_a} W^t_v \ge \sum_{j=t}^{t+\tau_a-1} D_{aj}$
$S_{at}=f_v X_{at}$
$\alpha_{vt} \leftarrow rnd\, U[0,1]$
$Y^q_{vt}=\alpha_{vt}+W^q_p$
state variable changes at each event